

Applied Precalculus

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Preface

As the title of the book suggests, our emphasis is on the real-life meaning of precalculus concepts. We want our students to master the basic concepts of precalculus while building a solid understanding of those concepts in practical contexts and the skills to apply mathematical tools to real-life processes.

To illustrate our approach, take for example the family of linear functions. When talking about linear functions, our emphasis is on the practical meaning and the units of the slope as well as on the units and the practical meaning of the horizontal and vertical intercepts. Such practical understanding is built through many applied examples and their careful analysis. We show how to use linear functions to model real-life processes which are described through tabulated data. At the same time, we try to ensure that students build the algebraic proficiency necessary to successfully work with linear functions.

We present all the main precalculus topics while stressing their practical meaning and their real-life applications. Those topics include:

- Functions, their graphs, and their numerical representation;
- Linear functions and their applications;
- Quadratic functions and quadratic equations;
- Algebra of exponential expressions, power functions, and their applications;
- Exponential and logarithmic functions, their algebraic properties and their rich variety of applications in the life sciences;
- Trigonometric functions and their applications to modeling periodic phenomena.

Our intended audience is primarily students in the life sciences and pharmacy. These students can be divided generally into two groups. The first group consists of students for whom this course is a terminal mathematics course, who wish to learn tools of before-calculus mathematics and to develop solid skills of

applying those tools to real-life problems. The second group consists of students who plan to continue their study of mathematics in an applied calculus course.

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Chapter 1

Introduction to Functions

The dose of medication that a pediatrician should prescribe to a child patient depends on the child's weight. The amount of medication remaining in a person's body depends on how much time has passed since they took the most recent dose. The distance between you and your destination depends on the speed at which you are traveling. The amount of practice problems you complete when studying for a mathematics class depends on how much time you spend working on them.

Each of the above describes a special type of dependence relationship between two variables called a *function*. Functions are used to model input—output relationships in a precise manner and are an essential part of the foundation for calculus and other advanced mathematical topics.

This chapter introduces the concept of a function and a variety of different ways that a function may be presented, including algebraic, graphical, and numerical representations. It also explains how functions are used to describe relationships in many real—world phenomena.

1.1 The Concept of a Function

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

One of the main concepts in mathematics and in the mathematical modeling of applied processes is that of a function. The formal mathematical definition of a function is abstract and general, but in applied precalculus we can think of a function as a special kind of dependence between two numerical variables.

Example 1.1.1

Suppose you are in Rhode Island and intend to buy a gift from an online store. The amount of money, C , that you will have to spend (that is, the total cost to purchase the gift) depends on the list price, p , of the item you select. The cost is equal to the list price plus the state of Rhode Island's 7% sales tax plus the rate the store charges for shipping and handling. Assuming that the online store charges a flat rate of \$4.95 for shipping and handling, give a formula for the cost C to purchase a gift of list price p dollars.

Solution. Given the list price p , you can calculate the cost C , in dollars, as

$$C = p + 0.07p + 4.95.$$

Here, $0.07p$ represents 7% of the list price—that is, the sales tax on the item.

In the above formula, C and p are variables—their values *vary* based on the item you select. They are both *numerical* variables as their possible values are numbers. The cost C depends on p and, moreover, the cost is uniquely determined by the item's list price p . We say in this case that C “is a function of p .” Using *function notation*, we write

$$C = f(p).$$

The notation $C = f(p)$ is read as “ C equals f of p .” Note that f does not stand for the word “function” here; it is merely a letter used to name the function and another letter could be used instead. The notation $f(p)$ represents the *value* of the function f at a given list price p ; that is, $f(p)$ is the *cost* corresponding to

a list price p :

$$f(p) = p + 0.07p + 4.95.$$

For every p , the value of the function $f(p)$ gives the cost C for a list price p . For example, $f(50)$ —the value of the function at $p = 50$ —gives the cost corresponding to the list price 50 dollars. Similarly, $f(60)$ gives the cost corresponding to the list price 60 dollars.

Example 1.1.2

Use the function

$$C = f(p) = p + 0.07p + 4.95$$

whose formula was found in Example ?? to determine the total amount you will spend to purchase a gift with a list price of $p = 80$ dollars from the online store.

Solution. We are looking for the cost C when $p = 80$. In other words, we are looking for $f(80)$ —read aloud as “ f of 80.” Using the formula for $f(p)$ with $p = 80$, we obtain

$$f(80) = 80 + 0.07 \cdot 80 + 4.95 = 90.55.$$

If you select an item listed at $p = 80$ dollars, you will pay $C = 90.55$ dollars. In function notation we could write $90.55 = f(80)$ or, equivalently, $f(80) = 90.55$.

Above, the cost variable, C , is called the *dependent variable* or the *output variable*. The list price, p , is the *independent variable* or the *input variable*. For each list price or input value—input for short—we have exactly one cost value. So we have exactly one output value—output for short—for each input. The formula for the function f gives the rule of how to obtain the output corresponding to each possible input p .

In general, we define a function as follows.

Definition 1.1.3 Informal Definition of a Function.

A function is a correspondence between two numerical variables, the input variable and the output variable, that takes input numbers from a certain subset of the real numbers called the *domain* of the function and prescribes to each ***exactly one*** output.

From the mathematical point of view, the domain of the formula for the function $f(p) = p + 0.07p + 4.95$ consists of all real numbers p . This is because $f(p)$ is unique and defined for each individual real number inputted for p . When we consider the applied context of this formula as presented in Example ?? and Example ??, however, the input p is a list price and therefore cannot be negative. For practical reasons, we would restrict the domain of f to positive inputs p .

Functions in General

In an abstract setting, when variables are not associated any specific real-life meaning, we commonly denote the independent variable by x , the dependent variable by y , and a function by f :

$$y = f(x).$$

Example 1.1.4

Let $y = f(x)$ where $f(x) = 0.2x + 1$.

1. What is the domain of f ?
2. Find $f(2)$ and $f(3)$.
3. Find all inputs x for which the output y is equal to 0.

Solution.

1. For every input number x , we can multiply x by 0.2 and add 1. So the output $f(x) = 0.2x + 1$ is defined and is a single number for each x . Hence, the domain of f consists of all real numbers x .
2. The quantity $f(2)$ is the value of the function or the output corresponding to the input $x = 2$. Substituting $x = 2$ into the formula for f , we find

$$f(2) = 0.2 \cdot 2 + 1 = 1.4.$$

Hence, the value of y corresponding to $x = 2$ is $y = 1.4$. Similarly,

$$f(3) = 0.2 \cdot 3 + 1 = 1.6.$$

3. We are looking for inputs x for which the output $y = 0$. In other words, we are looking for the value (or the values) of x which are solutions to the equation $0 = 0.2x + 1$ or equivalently $0.2x + 1 = 0$. To find x , we use the standard techniques of solving equations: we subtract 1 from both sides of the equation and then we divide both sides of the equation by 0.2. These steps are shown below.

$$\begin{aligned} 0.2x + 1 &= 0 \\ 0.2x + 1 - 1 &= 0 - 1 \\ \frac{0.2x}{0.2} &= \frac{-1}{0.2} \\ x &= -\frac{1}{0.2} \\ x &= -5 \end{aligned}$$

Functions in Applied Settings

The previous example illustrates that a function need not be associated with an applied scenario. When an application is involved, it is important to be able to interpret the practical meaning.

Example 1.1.5

The amount of nicotine in a person's bloodstream, $N = f(t)$, in milligrams, is a function of time t , in hours, that have passed since the person finished smoking a single cigarette.

1. Identify the independent variable and the dependent variable of this function.
2. In the statement $f(3) = 0.71$, what is the meaning of 3 and 0.71 in terms of time and nicotine? Include units in the answer.
3. Explain the meaning of the statement $f(4) = 0.5$ in practical terms.

Solution.

1. Here, N depends on t , making N a function of t . Thus N is the dependent variable and t is the independent variable.
2. The number 3 is a value of t , so its units are hours; 0.71 is a value of N , so its units are milligrams. In practical terms, $f(3) = 0.71$ means that 3 hours after a person finishes smoking a single cigarette there will be 0.71 milligrams of nicotine left in their bloodstream.
3. Similarly to the prior solution, $f(4) = 0.5$ means that 4 hours after a person finishes smoking a single cigarette there will be 0.5 milligrams of nicotine left in their bloodstream.

Example 1.1.6

In cases of strep throat, a daily pediatric dose of Amoxicillin, D , in milligrams, depends on the weight of a child, w , in kilograms; that is, D is a function of w . Denoting the function by g results in

$$D = g(w).$$

As a general rule, the daily dose should be 50 milligrams for each kilogram of weight.

1. Write a formula for the function $g(w)$.
2. Calculate the daily dose for a child who weighs 8 kilograms.

3. What does $g(11) = 550$ represent in practical terms?

Solution.

1. The value $g(w)$ represents the dose for the weight of w kilograms. Since the dose is 50 milligrams for each kilogram, the dose $g(w)$ is given by

$$50 \frac{\text{milligrams}}{\text{kilogram}} \times w \text{ kilograms.}$$

Multiplying and simplifying units results in $g(w) = 50w$ milligrams.

2. We calculate

$$g(8) = 50 \cdot 8 = 400$$

and arrive at the conclusion that the dose for a child who weighs 8 kilograms is 400 milligrams.

3. To answer questions like this, it can be useful to assign the appropriate units to each number. Since 11 is a value for w , it is measured in kilograms. 550 is a value for $g(w)$ or D , so it is measured in milligrams. Combining this information with the fact that $g(w)$ gives the dose for a child who weighs w kilograms, we conclude that the practical meaning of the statement $g(11) = 550$ is that a child weighing 11 kilograms should receive a daily dose of 550 milligrams of Amoxicillin when being treated for strep throat.

Example 1.1.7

A man leaves home and drives at 40 miles per hour toward a hospital located 150 miles away from his home. At time t hours after he began driving, his distance from the hospital, d , in miles, is given by

$$d = 150 - 40t.$$

1. Let $d = f(t)$. What is the formula for the function $f(t)$?
2. How far from the hospital is the man after 1.5 hours?
3. When will the man arrive at the hospital?

Solution.

1. Since $d = f(t)$ and the formula for d was given, $f(t) = 150 - 40t$.

2. The man's distance from the hospital at $t = 1.5$ is given by

$$f(1.5) = 150 - 40 \cdot 1.5 = 90.$$

After 1.5 hours, the man is 90 miles from the hospital.

3. The man will arrive at the hospital when his distance d from the hospital is 0 miles. We are looking for t such that $0 = 150 - 40t$ or equivalently $150 - 40t = 0$. Solving for t :

$$150 - 40t + 40t = 0 + 40t$$

$$150 = 40t$$

$$\frac{150}{40} = t$$

$$t = 3.75.$$

The man will arrive at the hospital 3 hours and 45 minutes after he left home.

Increasing and Decreasing Functions

In Example ?? we considered the function $D = g(w)$ that gives the daily dose D of Amoxicillin for a child weighing w kilograms. Notice that $g(w)$ increases as the weight w increases. In Example ??, the distance $d = f(t)$ from the hospital decreases as the number of hours t spent driving toward the hospital increases.

Definition 1.1.8 Increasing and Decreasing Functions.

- A function $f(x)$ is *increasing* if the values of $f(x)$ increase as x increases.
- A function $f(x)$ is *decreasing* if the values of $f(x)$ decrease as x increases.

Above, the dose of Amoxicillin would be an illustration of an increasing function of weight while the distance from the hospital would be an illustration of a decreasing function of time. We will see later that many functions are neither increasing nor decreasing through their domain; they may increase on some intervals and decrease on other intervals.

Functions as a Sequence of Operations

A function that is presented as a formula can be thought of as sequence of operations performed on an input in order to obtain the corresponding output. For example, the function $f(x) = 0.2x + 1$ of Example ?? takes an input, multiplies the input by 0.2, and then adds 1 to the result. The function performs this

sequence of operations no matter what the input is. We can write this symbolically as

$$f(\square) = 0.2 \cdot \square + 1.$$

Whatever input we feed into the function f , a number, an expression, whatever we like, the function will multiply the input by 0.2 and add 1.

By the way... In Chapter 7 we'll even consider putting *functions* into functions.

Example 1.1.9

Let $g(x) = 3 - 4x$. Evaluate and simplify each of the following.

1. $g(2 + h)$
2. $g(1 - h) + g(1)$

Solution. The function $g(x)$ takes any input $\square$ —whatever it might be—multiplies the input by 4, and subtracts the result from 3:

$$g(\square) = 3 - 4 \cdot \square.$$

1. In $g(2 + h)$, the input is an *algebraic expression*: $2 + h$. In any given context, h may stand for a number or for a numerical variable. No matter what h represents here, the function g takes the input $2 + h$, multiplies it by 4, and then subtracts the result from 3. Thus,

$$g(2 + h) = 3 - 4(2 + h).$$

To simplify, we use the distributive law to expand $4(2 + h)$:

$$g(2 + h) = 3 - 8 - 4h = -5 - 4h.$$

We cannot simplify further. The final answer is $g(2 + h) = -5 - 4h$.

2. We consider the whole expression all at once below, though you may find it helpful to separately find each of $g(1 - h)$ and $g(1)$ and then subtract the two results.

$$\begin{aligned} g(1 - h) + g(1) &= \left(3 - 4(1 - h)\right) + \left(3 - 4 \cdot 1\right) \\ &= \left(3 - 4 - 4(-h)\right) + \left(-1\right) \\ &= -1 + 4h - 1 \\ &= -2 + 4h \end{aligned}$$

Whatever h might be, $g(1 - h) + g(1) = -2 + 4h$.

Example 1.1.10

Find a formula for $f(x)$ if...

1. ... f takes the square root of the input, multiplies the result by 3, and subtracts 1.
2. ... f squares the input, adds 2, and takes the reciprocal of the result.

Solution.

1. We begin with an input x . Taking the square root of this input results in $\sqrt{x}$. Multiplying the result by 3 results in $3\sqrt{x}$. Finally, subtracting 1 results in $3\sqrt{x} - 1$. Therefore, the formula for f is

$$f(x) = 3\sqrt{x} - 1.$$

2. We begin with an input x . Squaring the input first results in x^2 . Adding 2 next results in $x^2 + 2$. Now, we take the reciprocal of $x^2 + 2$, which is $\frac{1}{x^2 + 2}$. Therefore, the formula for f is

$$f(x) = \frac{1}{x^2 + 2}.$$

Example 1.1.11

Find the domain of each of the following functions.

1. $f(x) = \frac{1}{x^2 - 1}$
2. $g(x) = \sqrt{x - 3}$

Solution. Note that neither $f(x)$ and $g(x)$ are assigned any specific practical applied meaning. This means that the domain of each function will be all inputs x for which the output is defined.

1. The function $f(x) = \frac{1}{x^2 - 1}$ is defined at any input that does not lead to division by zero and undefined at any input that does lead to division by zero. To find the value(s) of x for which $f(x)$ is undefined, we set its denominator equal to zero and solve for x :

$$\begin{aligned} x^2 - 1 &= 0 \\ x^2 - 1 + 1 &= 0 + 1 \\ x^2 &= 1 \end{aligned}$$

We can now see that $f(x)$ is undefined when $x = 1$ and when $x = -1$. The domain of $f(x)$ is all real numbers except for these two values. This can be written in a variety of different ways.

- Domain of $f(x) = \frac{1}{x^2 - 1}$ in words:

all real numbers $x \neq \pm 1$

- Domain of $f(x) = \frac{1}{x^2 - 1}$ in *set-builder notation*:

$$\{x \in \mathbb{R} : x \neq \pm 1\}$$

- Domain of $f(x) = \frac{1}{x^2 - 1}$ in *interval notation*:

By the way... The symbol $\cup$ is called a *union*. Essentially, it means “include all these intervals”.

$$(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$$

- The function $g(x) = \sqrt{x - 3}$ is defined provided the number under the radical is not negative. That is, it is defined for those values of x satisfying that $x - 3 \geq 0$. We can isolate x on one side of the inequality

$$\begin{aligned} x - 3 &\geq 0 \\ x - 3 + 3 &\geq 0 + 3 \\ x &\geq 3 \end{aligned}$$

which allows us to conclude that the domain of $g(x)$ is all real numbers x that are greater than or equal to 3. As before, this can be written in a variety of different ways.

- Domain of $g(x) = \sqrt{x - 3}$ in words:

all real numbers $x \geq 3$

- Domain of $g(x) = \sqrt{x - 3}$ in *set-builder notation*:

$$\{x \in \mathbb{R} : x \geq 3\}$$

- Domain of $g(x) = \sqrt{x - 3}$ in *interval notation*:

By the way... Notice that we wrote $[3, \infty)$ and not $(3, \infty)$ to denote that $x = 3$ is included in the domain.

$[3, \infty)$

Practice Problems

1.

a. $f(1)$

b. $f(-1.5)$

c. $f(0)$

2. The value of a car, V , in dollars, is a function of the number of years, t , since it was purchased; that is, $V = g(t)$.

The independent variable is $(\square \text{ } t \quad \square \text{ } V \quad \square \text{ } g)$ with units $(\square \text{ dollars} \quad \square \text{ years} \quad \square \text{ dollars/year})$.

The dependent variable is $(\square \text{ } t \quad \square \text{ } V \quad \square \text{ } g)$ with units $(\square \text{ dollars} \quad \square \text{ years} \quad \square \text{ dollars/year})$.

Exercises

1. The amount of caffeine, C , measured in milligrams, in a person's body t hours after drinking a cup of coffee is given by the function $C = f(t)$. What do each of the following statements tell you in practical terms; that is, in terms of time and caffeine? Answer in complete sentences and include units with each number.

(a) $f(0) = 96$

(b) $f(5) = 48$

(c) $f(24) \approx 0$

2. The total number of units, P , produced after t hours of production is given by the function $P = h(t)$. What do each of the following statements tell you in practical terms; that is, in terms of units of product produced and time? Answer in complete sentences and include units with each number.

(a) $h(1) = 10$

(b) $0 = h(0)$

(c) $h(3) \approx 29$

3. The total cost of a meal in a restaurant, C , in dollars, as a function of the menu price of the meal, p , also in dollars, is given by

$$C = p + 0.20p$$

where the term $0.20p$ corresponds to a 20% tip.

(a) What is the input variable and what units is it measured in?

(b) What is the output variable and what units is it measured in?

(c) Calculate the total cost of a meal whose menu price is \$25.

4. A bakery makes 6 batches of cookies an hour, with a batch typically consisting of 12 cookies. The total number of batches baked, B , is measured as function of time t , in hours, so that $B = f(t) = 6t$.

(a) What is the independent variable?

(b) What is the dependent variable?

(c) Evaluate $f(3)$. What is the practical meaning of $f(3)$ in the applied context of this problem?

5. A driver is heading to a faraway town. The amount of fuel, G , in gallons, left in the fuel tank is a function of the number of miles m driven during the trip; that is, $G = f(m)$.

(a) What does the statement $f(70) = 6$ tell you in practical terms? What are units of the numbers 70 and 6?

(b) What does the statement $f(200) = 1$ tell you in practical terms?

6. Let $f(x) = 4 - 3x$. Evaluate and simplify if possible:

(a) $f(3b)$

(b) $f(b + 1)$

(c) $f(b + 3) - f(3)$

7. Let $g(x) = 3x - x^2$. Evaluate:

(a) $g(2)$

(b) $g(-2)$

(c) $g(\sqrt{2})$

8. Let $g(x) = x^2 + 3$. Evaluate and simplify if possible:

(a) $g(h - 1)$

(b) $g(h + 1)$

(c) $g(b + 1) - g(1)$

9. Let $f(x) = \frac{1}{x + 1}$. Evaluate. If the value is undefined, say so.

(a) $f(0)$

(b) $f(-1)$

(c) $f(-2)$

10. Let $f(x) = \frac{3}{x + 2}$. Evaluate and simplify if possible.

(a) $f(h - 1)$

(b) $f(h + 1)$

(c) $f(b) - 4f(1)$

11. Let $f(t) = t^2 + \frac{1}{t}$. Evaluate and simplify if possible.

(a) $f(2)$

(b) $f(h + 1)$

(c) $f(h + 1) - f(2)$

12. Let $g(x) = 2x^2 + \frac{3}{2}x$ and $f(x) = x + 3$. Evaluate and simplify if possible.

(a) $f(2) + g(1)$

(b) $g(1) + 2f(x + 1)$

(c) $f(2x) - g(4 + x)$

13. A URI student tutors for MTH 103. The cost of each session in terms of hours, t , is given by the function $c(t) = 10t + 15$.

(a) What is the cost of a 3 hour session?

(b) If $c(t) = \$60$, how many hours was the session?

14. The value of a car, V , in dollars, t years after purchase is given by the function $V = g(t)$, where $g(t) = 16500 - 1500t$.
- (a) What is the value of the car at the time of purchase?
 - (b) What is the value of the car 5 years after purchase?
 - (c) After how many years is the car worth nothing?
15. Since you see lightning immediately and it takes the sound of thunder about 5 seconds to travel a mile, you can calculate the distance between you and the lightning. Count the number of seconds, S , between the flash of lightning and the sound of thunder. Then the distance, D , in miles, between you and the lightning is given by the function:

$$D = \frac{S}{5}$$

- (a) Identify the independent and the dependent variable.
 - (b) How far is the lightning if you counted 4 seconds between the flash and the thunder?
16. The cost of a medium pizza is \$10 plus \$1.75 per topping. Write a function $f(x)$ to describe the cost of a medium pizza with x number of toppings.
17. Find a formula for a function $h(x)$ for each of the following scenarios:
- (a) h multiplies the input by 7 then adds 5 to the result.
 - (b) h adds 5 to the input and then multiplies the result by 7.
18. Find a formula for a function $f(x)$ for each of the following scenarios:
- (a) f takes the square of the input, multiplies the result by 5, then subtracts 8.
 - (b) f multiplies the input by 5, takes the square of the result, then subtracts 8.
 - (c) f subtracts 8 from the input, takes the square of the result, then multiplies by 5.
19. Find a formula for a function $g(x)$ for each of the following scenarios:
- (a) g takes the input, divides it by 3, then adds this to the square root of the input times 4.
 - (b) g takes the reciprocal of the square root of the input, adds 1 to this, and then adds the input squared.
20. Find the domain of each of the following functions.
- (a) $f(x) = x - 1$

$$(b) \ f(x) = \frac{1}{x-1}$$

$$(c) \ f(x) = \frac{2}{x^2-4}$$

$$(d) \ f(x) = \sqrt{x}$$

$$(e) \ f(x) = x^2 - 4$$

21. Find the domain of each of the following functions.

$$(a) \ f(x) = x^n, \text{ where } n \text{ is a positive integer}$$

$$(b) \ f(x) = x^2 + x + \frac{2}{x}$$

$$(c) \ f(x) = \frac{1}{\sqrt{x}}$$